2	Tsiatsis, V., Karnouskos, S., Holler, J., Boyle, D., & Mulligan, C.(2018), *Internet of Things: technologies and applications for a new age of intelligence(1st Edition)*, Academic Press,ISBN: 978 - 0128144350
3	Pulver, T(2019), *Hands - On Internet of Things with MQTT: Build connected IoT devices with Arduino and MQ Telemetry Transport (MQTT)* (1st Edition) , Packt Publishing Ltd. ISBN: 9781789341782
4	Seneviratne, Pradeeka.(2018) *Hands - On Internet of Things with Blynk: Build on the power of Blynk to configure smart devices and build exciting IOT projects(1st Edition)*, Packt Publishing Ltd, ISBN: 9781788995061
5	Bahga, Arshdeep, and Vijay Madisetti (2014), *Internet of Things: A hands - on approach, Vpt (1st Edition)*, ISBN: 978 - 8173719547

Recommended by Board of Studies	24.04.2024
Approved by Academic Council	11.05.2024

Course Code	**Internet of Things Lab**	**L**	**T**	**P**	**C**
23CS2023		**0**	**0**	**3**	**1.5**

Course Objective:

Enable the students to:

1. Gain proficiency in iot devices, application areas, and associated technologies.
2. Recognize the iot sensors and technological challenges faced by iot devices.
3. Explore and learn iot with the help of Raspberry Pi and Arduino.

Course Outcomes:

The students will be able to:

1. Identify the building blocks of the Internet of Things and their characteristics.
2. Apply the concepts of iot.
3. Identify various sensors used for applications.
4. Demonstrate the working principles of sensors and Arduino boards.
5. Apply iot to various small - scale real - world applications.
6. Analyze the data received through sensors in iot.

Software Required:

Python - Arduino IDE

List of Exercises

1	Interfacing LED with an Arduino
2	Controlling an LED with a Push Button using Arduino
3	Interfacing a Temperature Sensor with an Arduino
4	Interfacing an LCD and a Temperature Sensor with an Arduino
5	Interfacing an Ultrasonic Sensor with an Arduino to Measure Distance
6	Interfacing an Ultrasonic Sensor and an LCD with an Arduino
7	Interfacing a DC Motor with an Arduino
8	Direction Control of a DC Motor Using an Arduino
9	Speed and Direction Control of a DC Motor Using an Arduino
10	Familiarization with Raspberry Pi/ Node MCU to perform necessary software installation and interface LED/Buzzer with Raspberry Pi.
11	Data Collection and Storage in Thingspeak Cloud using Raspberry Pi.
12	Data Analysis in Thingspeak Cloud.

Total Hours	**30 Hours**
Recommended by Board of Studies	**24.04.2024**
Approved by Academic Council	11.05.2024

Course Code	**Data Science Ecosystem**	**L**	**T**	**P**	**C**
23CS2024		**3**	**0**	**0**	**3**

Course Objectives

Enable the students to:

1. Identify the fundamental principles of data science and its practical uses.
2. Implement simple applications and analyze the results using relevant tools.

3. Establish ethical guidelines for data science projects and ensure accountability in decision - making processes.

Course Outcomes:

The students will be able to:

1. Illustrate the fundamental concepts and purposes of data science.
2. Describe the processes involved in working with data.
3. Apply descriptive statistics and exploratory data analysis techniques.
4. Recognize different types of machine learning algorithms and their applications.
5. Utilize content - based and collaborative filtering techniques to develop recommender systems.
6. Infer the ethical considerations in data science.

Module: 1	**Basics of Data Science**	**7 Hours**
Introduction to Data Science: Applications of data science - Properties of Data: Exploring various datasets in different repositories - Tool Boxes for Data Scientists.		

Module: 2	**Understanding Data**	**8 Hours**
Working with Data: Import – Select – Filter - Manipulate - sort - group - rearrange - rank and analyze the data for missing data values. Data visualization: Plot various plots for the given dataset.		

Module: 3	**Statistical Inference**	**8 Hours**
Descriptive statistics - Exploratory Data Analysis: Calculate the mean - median - variance - standard deviation for the given small and large dataset - analyze the correlation between the variables in the dataset - Estimation - Hypothesis Testing: Formulate Null and Alternative hypotheses for real - world use cases.		

Module: 4	**Machine Learning**	**8 Hours**
Introduction to machine learning - Types of machine learning - Linear - Multiple - Logistic and Polynomial Regression. KNN - Decision Trees. Clustering - Similarity and Distance measure - K means clustering. Agglomerative Clustering. Model evaluation.		

Module: 5	**Recommender System**	**7 Hours**
Content - Based Filtering - Collaborative Filtering: Developing a retail recommendation system - Hybrid Recommenders: Hotel recommendation system - Evaluating Recommenders.		

Module: 6	**Ethics in Data science**	**7 Hours**
Ethical Frameworks and Principles - Concepts of Privacy and Confidentiality: Ethical Issues in Data Collection and Consent - Techniques for Anonymization and De - identification - Bias and Fairness: basics - Fairness in Machine Learning Models - Transparency and Accountability: Importance of Transparency in Data Science - Explainability and Interpretability of Models - Establishing Accountability in Data Science Practices		
Total Lectures		**45 Hours**

Text Books

1. Laura Igual, Santi Seguí (2017), *Introduction to Data Science: A Python Approach to Concepts, Techniques and Applications(1st edition)*, Springer, ISBN: 978-3-319-500171.
2. David Martens(2022), *Data Science Ethics: Concepts, Techniques and Cautionary Tales (1st Edition)*, Oxford University Press, ISBN: 978-0-19-284726-3.

Reference Books

1. Steven Cooper(2018), *Data Science from Scratch: The #1 Data Science Guide for Everything A Data Scientist Needs to Know: Python, Linear Algebra, Statistics, Coding, Applications, Neural Networks, and Decision Tree (1st Edition)*, ISBN-13: 978-1723141201.
2. Cathy O'Neil and Rachel Schutt (2014), *Doing Data Science, Straight Talk from The Frontline (1st Edition)*, O'Reilly, ISBN: 978-1-449-35865-5.
3. Sinan Ozdemir(2016), *Principles of Data Science (1st Edition)*, Packt Publishing Limited, ISBN: 978-1785887918
4. V. K. Jain (2018), *Data Science and Analytics (1st Edition)*, Khanna Publishing, ISBN - 13: 978-9386173676.
5. Jake Vander Plas(2016), *Python Data Science Handbook: Essential Tools for Working with Date,*

	Shroff/O'Reilly, ISBN: 978-9352134915
6	Peter Morgan (2018), *Data Science from Scratch with Python: Step - By - Step Guide (*2nd edition*)*, Createspace Independent Publishing Platform, ISBN: 978-1726020688.
Recommended by Board of Studies	24.04.2024
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Course Code	**Data Science Ecosystem Lab**	**L**	**T**	**P**	**C**
23CS2025		**0**	**0**	**3**	**1.5**

Course Objectives:

Enable the students to:

1. Develop expertise in data manipulation, exploratory analysis, and decision - making for comprehensive data management and analysis
2. Acquire proficiency in neo4j, python libraries, tableau, and regression technique performance analysis.
3. Illustrate analysis on classification, tree - based methods and skilled in nlp techniques

Course Outcomes:

The students will be able to:

1. Express proficiency in data manipulation with Pandas and Numpy.
2. Develop expertise in exploratory data analysis and statistical inference techniques for robust data - driven decision - making.
3. Represent proficiency in working with Neo4j for graph database management and data visualization using Python libraries.
4. Create visualization dashboards with Tableau and perform regression technique performance analysis.
5. Illustrate performance analysis on classification techniques, with a focus on tree - based classification methods
6. Develop proficiency in working with NLP techniques and acquiring/analyzing data from real - time systems.

List of Exercises

1. Introduction to Data Science with Python programming
2. Working with Data using Pandas, Numpy
3. Exploratory Data analysis
4. Statistical Inference
5. Working with Neo4j
6. Data visualization using Python Libraries
7. Visualization Dashboard using Tableau
8. Performance analysis on regression techniques
9. Performance analysis on classification techniques
10. Performance analysis of tree based classification techniques
11. Working with NLP
12. Data acquisition and analysis from real time systems.

Total Hours	**30 Hours**
Recommended by Board of Studies	24.04.2024
Approved by Academic Council	11.05.2024

Course Code	**Machine Learning Techniques**	**L**	**T**	**P**	**C**
23CS2028		**3**	**0**	**0**	**3**

Course Objectives:

Enable the students to:

1. Recognize the human learning aspects and primitives in the learning process facilitated by computers.
2. Analyze the nature of problems solved with machine learning techniques.
3. Design and implement suitable machine learning technique for a given application.

Course Outcomes:

The students will be able to:

1. Recall the concepts and background of machine learning techniques.